United International University

Assignment

“Designing an Enterprise Network using Packet Tracer”

Submitted To:

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Course: **Computer Networks Laboratory**
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Introduction

In this project, I designed and implemented an enterprise network for UIU using Cisco Packet Tracer. The network consists of **30 hosts** (including PCs, laptops, and servers), **7 routers**, and **8 switches.** The network is divided into a private and a public network, connected using NAT (Network Address Translation). I used DHCP for IP address assignment, configured routing protocols (RIP), and ensured that the network was fully operational by testing connectivity between all devices. The design mimics a real-world enterprise network, with organized subnets and clean device placement.

IOS Commands Used

Below are the key IOS commands used to set up the **R3 router** and configure the network:

Basic Router Configuration for interface gig0/0

```
Router>
Router>en
Router#configure terminal
Router(config)#interface gig0/0
Router(config-if)#ip address  172.16.0.18  255.255.255.240
Router(config-if)#exit
Router(config)#exit
router#  wr
```

DHCP Configuration

```
Router(config)# ip dhcp pool LAN2
Router(dhcp-config)# network 172.16.0.17  255.255.255.240
Router(dhcp-config)# default-router 172.16.0.18
Router(dhcp-config)# dns-server 8.8.8.8

(rest)
```

RIP Configuration

```
Router(config)# router rip
Router(config-router)# version 2
```

```
Router(config-router)# network 172.16.0.17
(rest)
```

NAT Configuration

```
Router(config)# interface gig0/0
Router(config-if)# ip nat inside
Router(config-if)# interface gig0/1
Router(config-if)# ip nat outside

(rest)
```

```
Router(config)# ip nat inside source static 172.16.0.19 100.20.10.5

(rest)
```

Challenges Encountered

a. Router Configuration

Initially, there were challenges in configuring RIP on all routers to ensure proper propagation of routes across the network.

b. NAT Implementation

I implemented static NAT to connect the private and public networks. Assigning public IPs to all devices would have been inefficient, so NAT was chosen to simplify this process. However, despite attempting the configuration multiple times, NAT did not function as expected. Challenges were faced in ensuring proper traffic translation, even after careful testing and verification of the setup.

Summary of Learning

Through this project, I gained hands-on experience in designing and configuring an enterprise network using Cisco Packet Tracer. Key takeaways include:

- Understanding routing protocols such as RIP and their implementation.
- Configuring NAT for network segmentation and secure communication.

- Using DHCP for automated IP address assignment.

The project enhanced my practical skills in network design and configuration, preparing me for real-world networking scenarios.